# 2.1 The structure of a data packet

The amount of data that is stored in a file can be very large. If you tried to transmit this data from one device to another, all at once, this just wouldn't be possible or practical. The wires or radio waves used would simply not be able to accommodate sending such large amounts of data at a single time. Therefore, for the data in a file to be transmitted, it is broken down into very small units called **packets**.

Each packet of data contains three different sections:

| Packet header |
|---|
| Payload |
| Trailer |

The **packet header** contains a lot of important information about the data enclosed in the packet and its destination. The information it includes is the:

- destination address
- packet number
- originators address.

The destination address is normally an **IP address**. It is the IP address of the device where the data is being sent. Without this address, the hardware transmitting the data would not know where to send the data.

Each data packet in the file is given a packet number. The packets of data may not have all been sent in the correct order, this will depend on the type of transmission used; you will learn more about this in Sections 2.2 and 2.3. This means that when the destination device has received all the data packets, it can use the packet number to put them back into the correct order to recreate the file.

The originators address is also normally an IP address. It is the IP address of the device from which the data has been originally sent. So, if you send data to another device, the originator's address is the address of your device. This address isn't crucial to the transmission process, but it does help to trace where data has been sent from in situations such as illegal activity, or to simply request the original device to resend the data if an error is detected.

The **payload** of the data packet is the actual data from the file that you are sending. The data is broken up into many small units to be sent as the payload in each packet.

The **trailer** section of the packet is sometimes known as the footer. This contains two main pieces of information. The first is the marker to indicate it is the end of the packet and also the data for any error detection systems that are being used. You will learn more about this in Section 2.2.

**KEY WORDS**

**packet:** a small unit of data.

**packet header:** a section of a packet of data that contains information about the contents of the packet and its destination.

**internet protocol (IP) address:** the unique address that is given to a device when it is connected to a network.

**KEY WORDS**

**payload:** the actual data that the user is sending to the receiver.

**trailer:** a section of a packet of data that contains information about any error checking methods that may be used.

## Questions

1 Can you name the **three** main sections of a data packet?
2 What information does the packet header contain?
3 Which part of the data packet is the actual data from the file?
4 Why does data need to be broken down into packets to be transmitted?

## ACTIVITY 2.1

Ask your teacher if you are allowed to use a packet tracer. Your teacher may want to complete this activity together as a class. Use the packet tracer to send data to another device and look at the data that the packet tracer software provides. What kind of information does it give you? What route did your packets take to get to the destination device?

# 2.2 Packet switching

You have learnt that data is broken down into packets to be sent from one device to another. The process of transmitting these packets is called **packet switching**. In this process, each packet of data is sent individually from one device to another. There could be several pathways across a **network** that the packets could be transmitted. Each data packet could be sent along a different pathway. A **router** is the device that controls which pathway will be used to transmit each packet. You will learn more about how a router does this later in the chapter. Look at the diagram of a network in Figure 2.2:

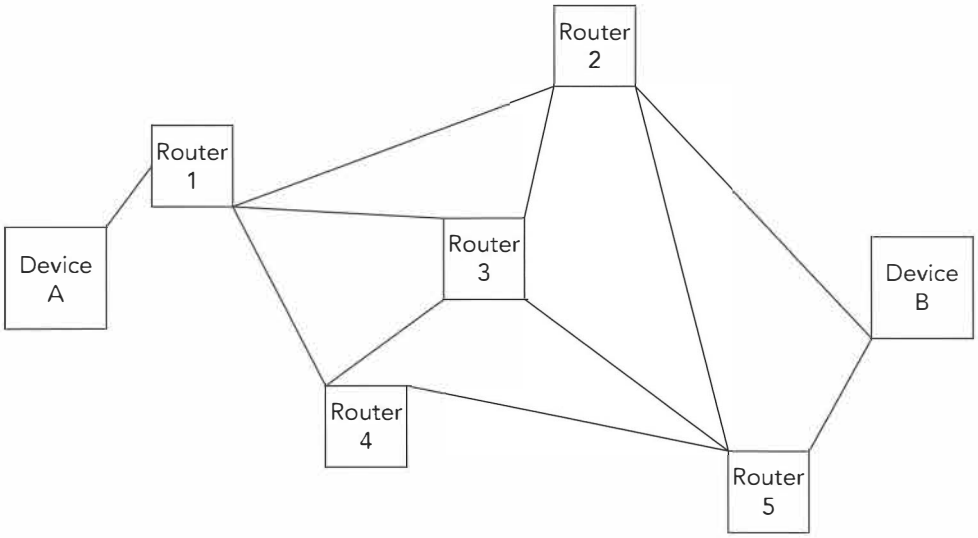



**Figure 2.2:** A network

The packets of data start at device A, they could be transmitted down any of the pathways between device A and device B. Each packet can be sent using a different pathway. When a packet reaches a router, the router decides which pathway to send the packet along next. This will continue until all the packets have arrived at device B. It is likely that the packets will arrive out of order. Once all packets have arrived, they are reordered to recreate the file.

## KEY WORDS

**packet switching:** a method of transmitting data packets across a network. Each data packet is able to take an individual pathway across the network.

**network:** computers and devices that are joined together using cables or wireless technology.

**router:** a network component that examines a data packet to obtain its destination address and then forward the packet to this address.

## ACTIVITY 2.2

Five packets of data need to be sent from device A to device B.

The pathway the first packet takes is:

- Router 1
- Router 2.

The pathway the second packet takes is:

- Router 1
- Router 4
- Router 5.

The pathway the third packet takes is:

- Router 1
- Router 3
- Router 2.

The pathway the fourth packet takes is:

- Router 1
- Router 4
- Router 3
- Router 5.

The pathway the fifth packet takes is:

- Router 1
- Router 3
- Router 5.

Draw a copy of the diagram of a network. Use coloured pens to draw the pathway that each of the packets of data takes.

### Self Assessment

What did you do to check if your diagram is correct? Have you tried tracing each route to see if it goes through the correct routers?

# 2.3 Methods of data transmission

You might think that there is only one method that can be used to transmit data from one device to another, but there are actually several. You need to understand how data is transmitted using each of these methods. These include:

- serial
- parallel
- simplex
- half-duplex
- full-duplex.

**Serial** and **parallel** data transmission involve the number of wires that are used to transmit the data, and how many bits of data are sent at a time.

KEY WORDS

**serial:** a transmission method where data is sent one bit at a time down a single wire.

**parallel:** a transmission method where data is sent multiple bits at a time down multiple wires.

## Serial data transmission

In serial data transmission, data is transmitted using a single wire. Each bit of data is transmitted one at a time along the single wire. Figure 2.3 shows an example:

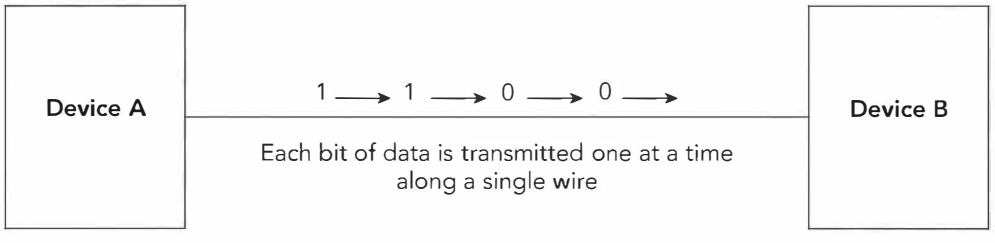



**Figure 2.3:** Serial data transmission

There are several advantages and disadvantages of serial data transmission, see Table 2.1:

| Advantages | Disadvantages |
|---|---|
| As data is sent one bit at a time, it should arrive in order of sequence. This means there is less chance of the data being skewed. | As data is sent one bit at a time, the transmission of data is slower. |
| As data is sent along a single wire, there is less chance of **interference**. This means there is less chance of error in the data. | Additional data may need to be sent to indicate to the receiving device when the data transmission has started and stopped. These are called a start bit and a stop bit. |
| Only one wire is needed for a serial transmission cable, therefore, it is cheaper to manufacture and also cheaper to buy. | |

**Table 2.1:** Advantages/disadvantages of serial data transmission

KEY WORD

**interference:** disruption, such as electromagnetism, to data when it is transmitted.

Serial transmission has less chance of data being skewed and less chance of interference. This makes it more suitable for the transmission of data over long distances.

## Parallel data transmission

In parallel data transmission, data is transmitted using multiple wires. Multiple bits of data are transmitted along each wire at the same time. Figure 2.4 shows an example:

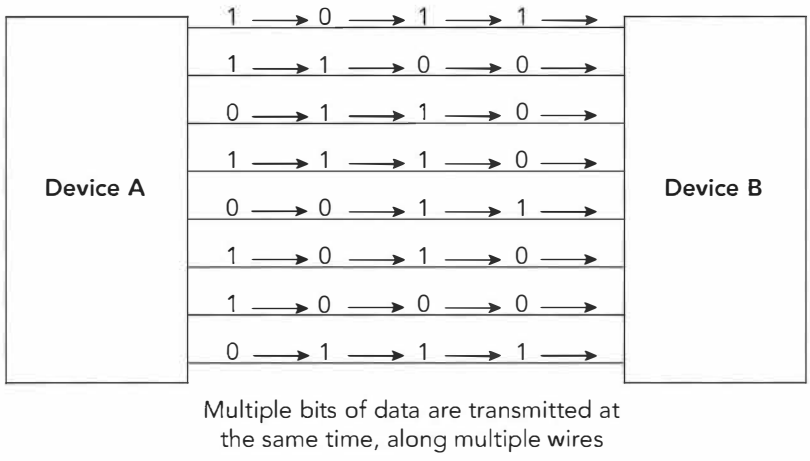



**Figure 2.4:** Parallel data transmission

There are several advantages and disadvantages of parallel data transmission, see Table 2.2:

| Advantages | Disadvantages |
|---|---|
| As data is sent multiple bits at the same time, the transmission of data is quicker. | As data is sent multiple bits at the same time, bits do not arrive in order and need to be reordered after transmission. This increases the risk of the data being skewed. |
| Many computers and devices use parallel data transmission to transmit data internally. Therefore, there is no requirement to convert this to serial data transmission to transmit the data across a network. | As data is sent along multiple wires, there is more chance of interference. This means there is more chance of error in the data. |
| | Multiple wires are needed for a parallel transmission cable, therefore, it is more expensive to manufacture and also more expensive to buy. |

**Table 2.2:** Advantages/disadvantages of parallel data transmission

Parallel transmission has an increased chance of data being skewed and an increased chance of interference. This makes it more suitable for the transmission of data over shorter distances. This is often limited to approximately 5 metres.

## Question

5 A marketing company prints leaflets to deliver to people's houses. The leaflets are designed on a computer and sent to a large printer that is 3 metres away. Large numbers of leaflets need to be printed in a short period of time, to makes sure that they are ready for delivery. Would you use a serial or a parallel data transmission cable to connect the computer to the printer? Why would you choose this type of cable?

**Simplex**, **half-duplex** and **full-duplex** transmission involve the direction in which the data is transmitted.

## Simplex data transmission

In simplex data transmission, data is transmitted from one device to another in one direction only. Figure 2.5 shows an example:




**Figure 2.5:** Simplex data transmission

You would only use this type of data transmission when there is no need to send data both ways between the devices. One example is connecting a keyboard to a computer. Simplex data transmission can be used to do this as data only needs to be sent from the keyboard to the computer, so that the computer screen can display the keys that are pressed. There is no reason why the computer would need to send data to the keyboard, so a simplex data transmission connection is used.

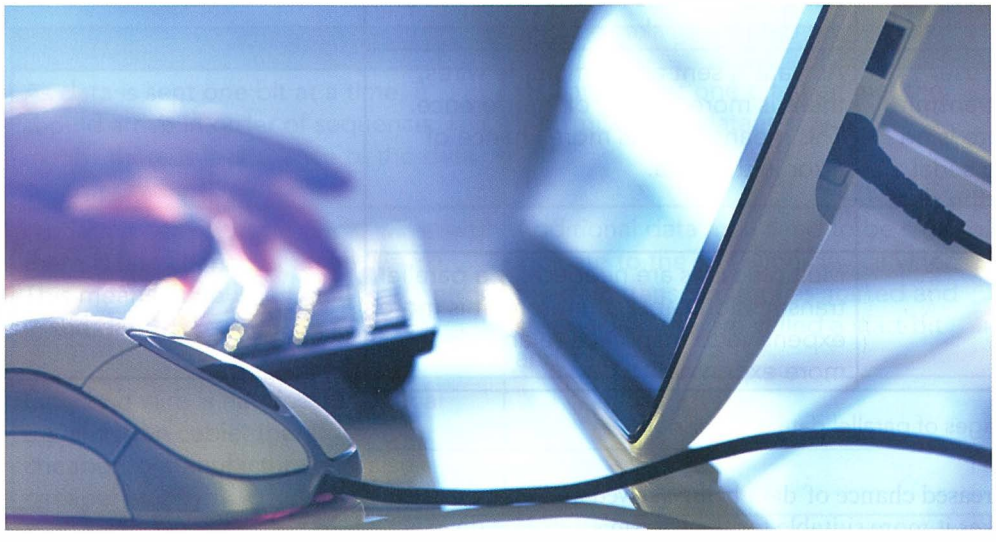


**Figure 2.6:** Simplex data transmission is used between a computer and a keyboard

**KEY WORDS**

**simplex:** a transmission method where data is transmitted in a single direction only.

**half-duplex:** a transmission method where data is transmitted in both directions, but only one direction at a time.

**full-duplex:** a transmission method where data is transmitted in both directions at the same time.

# Half-duplex data transmission

In half-duplex data transmission, data can be transmitted in both directions between the devices, but only one direction at a time. Figure 2.7 shows an example:

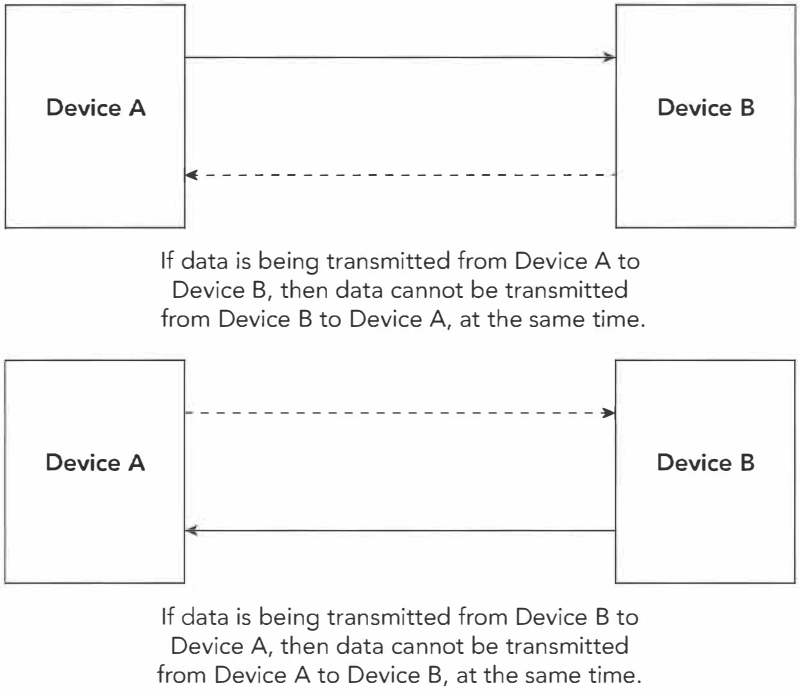



**Figure 2.7:** Half-duplex data transmission

You would use this type of data transmission when you need to send data in both directions between the devices, but there is no requirement for this to be at the same time. It can also be used when the communication between two devices needs to have a high level of performance. This is when two half-duplex connections can be set up between devices to allow data to be transmitted in both directions at the same time, but using two different channels of communication. Can you work out how this would be set up?

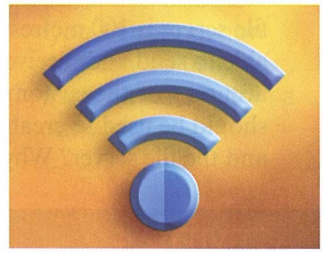


**Figure 2.8:** Wi-Fi uses half-duplex data transmission

## ACTIVITY 2.3

Use the internet to research how half-duplex data transmission is used to create a Wi-Fi connection. Why is half-duplex, rather than full-duplex data transmission used in Wi-Fi connections?

# Full-duplex data transmission

In full-duplex data transmission, data can be transmitted in both directions between the devices, at the same time. Figure 2.9 shows an example:

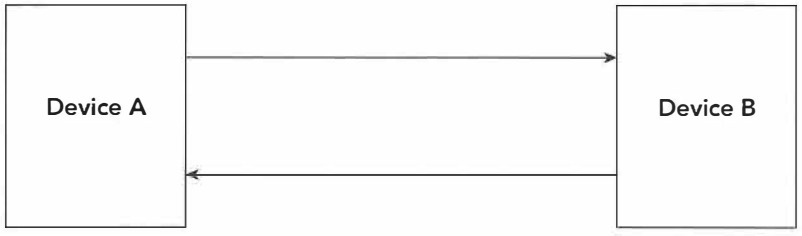



**Figure 2.9:** Full-duplex data transmission

You would use this type of data transmission when it is essential for data to be both sent and received by each device, at the same time. One example is a telephone conversation. The person speaking into Device A can speak at the same time as the person speaking into Device B. It may be a little bit of a chaotic conversation if you both try and speak at the same time all the time though! However, during most conversations, people do sometimes speak at the same time, so full-duplex data transmission is needed.

When a connection is created between two different devices it will either be a serial or parallel data connection. It will also be a simplex, half-duplex or full-duplex connection. This means that a connection can be, for example, a serial simplex connection, or a parallel half-duplex connection, or possibly a serial full-duplex connection. You need to understand when each of the different methods of transmission are most suitable.

## Questions

6 Which method of data transmission sends data along a single wire?

7 Which method of data transmission sends data multiple bits at a time?

8 If a data transmission connection sends data one bit at a time, in both directions, but not at the same time, what kind of data transmission methods are used?

9 What are **two** advantages of serial data transmission?

10 What are **two** disadvantages of parallel data transmission?

11 A business manager transmits data about its customers to a central file server. The file server is 100 metres away from the business manager's office. They need to be able to send and receive customer data to and from the server, at the same time. The accuracy of the customer's data is imperative. Which methods of data transmission should be used to create the connection between the business manager's computer and the file server? Why would you choose those methods?

### ACTIVITY 2.4

Write a data transmission scenario like the one given in Question 11. Think about which data transmission methods could be used for your scenario and why they would be the most suitable.

#### Peer Assessment

Give the scenario to a friend and ask them which data transmission methods they would use. Are they the same methods that you thought should be used? If you have chosen different methods, discuss and come to an agreement about which methods should be used. If you chose the same methods, see if you had the same reasons for choosing those methods.